

NEW PRODUCTS

MAKE IN INDIA

COMPONENTS & CHIPS

62mm CoolSiC module

Infineon Technologies has expanded its CoolSiC 1200V and 2000V MOSFET module families with the introduction of a 62mm industry-standard package. M1H technology broadens the gate voltage window, improving robustness



against voltage spikes at high switching frequencies and enabling

SiC use in mid-power applications from 250kW. The expanded range caters to sectors such as solar energy, servers, energy storage, electric vehicle chargers, traction, commercial induction cooking, and various power conversion systems. The module has minimal switching and transmission losses, reducing the need for extensive cooling systems. It has high thermal cycling capacity and a continuous operating temperature of 150°C.

Infineon Technologies
<https://www.infineon.com>

1200V MOSFETs

SemiQ has launched a family of 1200V MOSFETs, available with or without 1200V SiC Schottky diodes in a SOT-227 package. The SiC modules, made



from high-performance ceramics, are engineered to provide reliable performance even in harsh

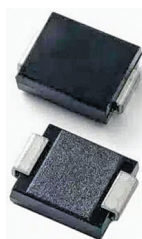
conditions. These modules feature high breakdown voltage, capability for high-temperature operation, low $R_{ds(On)}$ shift over a wide temperature

range, and excellent gate oxide stability and ruggedness. They are suitable for use in electric vehicle charging, DC-DC converters, energy storage systems, and power supplies across sectors such as medical, solar, wind energy, and data centres. Designed with low switching losses and minimal thermal resistance from junction to case, these modules ensure efficient operation and are ruggedly built for easy mounting.

SemiQ
<https://semi.com>

Surge protection thyristors

Littelfuse, Inc. has unveiled the Pxxx0S3N-A Automotive Grade 3kA SIDACtor Protection Thyristor Series, now available in the DO-214AB package. The distinguishing feature of the series is its high power density design,



capable of withstanding surge currents of up to 3kA (8/20µs) and voltages up to 6kV while maintaining a low clamping voltage. Specifically engineered for applications in battery

electric vehicles and plug-in hybrid electric vehicles, the series delivers unparalleled transient surge protection for onboard chargers when used in tandem with an MOV combination. This series promises enhanced protection for EVs and energy storage. Its compact form factor and surge rating make it suitable for highly exposed applications.

Littelfuse, Inc.
<https://www.littelfuse.com>

Rectifier and TVS 2-in-1

New Yorker Electronics has introduced Vishay's R3T2FPHM3. This device marks the industry's first-ever combination of a standard rectifier and transient voltage suppressor (TVS) in a single unit. Designed to



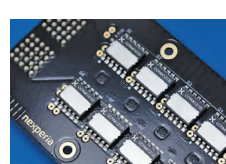
address the automotive sector's space-saving needs, the R3T2FPHM3 boasts an oxide

planar chip junction design and a standard cathode circuit configuration. It provides secondary protection for sensor units, supports distributed airbag modules, and facilitates low-power DC/DC converters within power distribution systems. It can serve as a secondary protective measure for sensor units, ensuring their operation. Additionally, it finds valuable utility in supporting distributed airbag modules, enhancing safety features in automotive systems.

Vishay
<https://www.vishay.com>

High-voltage GaN FET devices

Nexperia has launched gallium nitride (GaN) field-effect transistor (FET) devices utilising high-voltage GaN high-electron-mobility transistor (HEMT) technology and proprietary copper-



clip CCPAK surface mount packaging. The GAN039-650NTB GaN FET, housed in

CCPAK1212i top-side cooling packaging, offers benefits for renewable energy applications like solar and residential heat pumps, as well as various